

10-606 Recitation

Justin

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Warmup

Solve the following matrix equation for X : $(A^\top XB)^{-1} + C = D$

Solution:

$$(A^\top XB)^{-1} = D - C$$

$$A^\top XB = (D - C)^{-1}$$

$$(A^\top)^{-1}A^\top XB = (A^\top)^{-1}(D - C)^{-1}$$

$$XB = (A^\top)^{-1}(D - C)^{-1}$$

$$XBB^{-1} = (A^\top)^{-1}(D - C)^{-1}B^{-1}$$

$$X = (A^\top)^{-1}(D - C)^{-1}B^{-1}$$

PSD Matrices and Inverses

1. Show that if A is invertible, then its eigenvalues are all nonzero. Conversely, if an eigenvalue is zero, why can't A^{-1} exist?

Solution: If $Av = \lambda v$ and A^{-1} exists, then $v = A^{-1}Av = Iv = A^{-1}\lambda v$. Both A and v are non-zero so $\lambda \neq 0$. Conversely, if $\lambda = 0$, then $Av = 0$ for $v \neq 0$. Since A is zero, its determinant is also zero meaning A must be singular and non-invertible.

2. If A is invertible, verify that

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}u v^\top A^{-1}}{1 + v^\top A^{-1}u},$$

as long as the denominator is nonzero.

Solution: Let $s = 1 + v^\top A^{-1}u$ be the denominator (what are its dimensions?). First, we multiply both sides by $(A + uv^\top)$. This gives us I

on the left side and, on the right side, we get:

$$\begin{aligned}
& A^{-1}(A + uv^{\top}) - \frac{1}{s}(A^{-1}uv^{\top}A^{-1})(A + uv^{\top}) \\
&= A^{-1}A + A^{-1}uv^{\top} - \frac{1}{s}(A^{-1}uv^{\top}A^{-1}A + A^{-1}uv^{\top}A^{-1}uv^{\top}) \\
&= I + A^{-1}uv^{\top} - \frac{1}{s}(A^{-1}uv^{\top} + A^{-1}uv^{\top}A^{-1}uv^{\top}) \\
&= I + A^{-1}uv^{\top} - \frac{A^{-1}u}{s}(I + v^{\top}A^{-1}u)v^{\top} \\
&= I + A^{-1}uv^{\top} - \frac{A^{-1}u}{s}sv^{\top} \\
&= I + A^{-1}uv^{\top} - A^{-1}uv^{\top} \\
&= I.
\end{aligned}$$

3. A *norm* is a function ρ that satisfies

- (i) $\rho(x) \geq 0$ for all x
- (ii) $\rho(x) = 0$ if and only if $x = 0$
- (iii) $\rho(cx) = |c|\rho(x)$ for all x and scalars c
- (iv) $\rho(x + y) \leq \rho(x) + \rho(y)$ for all x, y .

If A is a positive-definite matrix A , show that $\rho(x) = \|x\|_A$ where $\|x\|_A = \sqrt{x^{\top}Ax}$ defines a norm. Recall that A is positive-definite if $x^{\top}Ax > 0$ for all nonzero x . Hint: You may use the Cauchy-Schwarz inequality $x^{\top}Ay \leq \|x\|_A\|y\|_A$ without proof.

Solution: (i) and (ii) are immediately implied by the fact that A is positive definite. For (iii), $\rho(cx) = \|cx\|_A = \sqrt{(cx)^{\top}A(cx)} = \sqrt{c^2x^{\top}Ax} = |c|\sqrt{x^{\top}Ax} = |c|\|x\|_A = |c|\rho(x)$ as desired. For (iv), we will show that $\|x + y\|_A^2 \leq (\|x\|_A + \|y\|_A)^2$. We have

$$\begin{aligned}
\|x + y\|_A^2 &= (x + y)^{\top}A(x + y) = x^{\top}Ax + x^{\top}Ay + y^{\top}Ax + y^{\top}Ay \\
&= x^{\top}Ax + 2x^{\top}Ay + y^{\top}Ay \\
&= \|x\|_A^2 + 2x^{\top}Ay + \|y\|_A^2.
\end{aligned}$$

Since A is PSD, it must also be symmetric meaning $A^{\top} = A$. Then, $(x^{\top}Ay)^{\top} = y^{\top}A^{\top}x = y^{\top}Ax$ so $x^{\top}Ay = y^{\top}Ax$ and we can write $x^{\top}Ay + y^{\top}Ax = 2x^{\top}Ay$ or $2y^{\top}Ax$.

On the right-hand side,

$$(\|x\|_A + \|y\|_A)^2 = \|x\|_A^2 + \|y\|_A^2 + 2\|x\|_A\|y\|_A.$$

So,

$$\|x\|_A^2 + 2x^{\top}Ay + \|y\|_A^2 \leq \|x\|_A^2 + \|y\|_A^2 + 2\|x\|_A\|y\|_A$$

$$x^T A y \leq \|x\|_A \|y\|_A$$

From here the desired inequality follows after applying Cauchy-Schwarz.

SVD and Rank

1. What's the singular value decomposition of a matrix A ?

Solution: For any $A \in \mathbb{R}^{m \times n}$, there exists some $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ orthogonal and $\Sigma \in \mathbb{R}^{m \times n}$ diagonal (and nonnegative) such that $A = U \Sigma V^T$.

2. Show that if $A = uv^T$ with $u \in \mathbb{R}^m$, $v \in \mathbb{R}^n$ and $u, v \neq 0$, then $\text{rank}(A) = 1$.

Solution: $\text{col}(A) = \text{span}\{u\}$ since each column j equals $v_j u$. Hence $\text{rank} = 1$.

3. What is the relation between singular values of A and eigenvalues of $A^T A$?

Solution: The singular values of A are the square roots of the eigenvalues of the matrix $A^T A$.

$A^T A = (U \Sigma V^T)^T (U \Sigma V^T) = (V \Sigma^T U^T) (U \Sigma V^T) = V (\Sigma^T \Sigma) V^T$ because U is orthogonal so $U^T U = I$. Since Σ is diagonal, $\Sigma^T = \Sigma$ which means that the eigenvalues of $A^T A$ are $\sigma_1^2, \dots, \sigma_n^2$, which are the squares of the singular values of A .

4. Compute the nonzero singular value of $A = uv^T$.

Solution: Use the logic implied by the previous question. We have $A^T A = vu^T uv^T = \|u\|^2 vv^T$. Eigenvalues and eigenvectors of $A^T A$ thus satisfy $\|u\|^2 vv^T x = \lambda x$. Then,

$$(A^T A)v = (\|u\|^2 vv^T)v = \|u\|^2 v(v^T v) = \|u\|^2 v \|v\|^2 \text{ where } \lambda = \|u\|^2 \|v\|^2.$$

Now, since vv^T has rank one, we know that this must be the only nonzero eigenvalue. From here,

$$\sigma(A) = \sqrt{\lambda} = \|u\| \|v\|.$$

5. If $A = U \Sigma V^T$, what is the SVD of A^T ?

Solution: $A^T = V \Sigma^T U^T$ with the same nonzero singular values and swapped singular vectors.